

# Tianyi Ouyang

Ph.D.in Physics

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## RESEARCH INTERESTS

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Excitonic and electronic properties in two-dimensional semiconductor materials (mainly Transition Metal Dichalcogenides); Optical spectroscopy; Physical modelling

## EDUCATION

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**University of California, Riverside**, Riverside, California, USA September 2019 to June 2025

Ph.D. in Physics

Advisors: Chun Hung Lui (Chair); Nathaniel M. Gabor (Committee); Yong-Tao Cui (Committee)

**University of Science and Technology of China**, Hefei, Anhui, China August 2015 to June 2019

B.S. in Physics (Theoretical physics)

Advisor: Zhenyu Zhang

## AWARD

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**Dissertation Completion Fellowship Award** Jan 2025 to June 2025

**Outstanding Teaching Assistant Award** Sept 2024 to June 2025

**USTC Outstanding Student Scholarship Silver Award** 2016

## CONFERENCE

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**Presenter in American Physical Society March Meeting 2025** March 2025

Enhancing resonant second harmonic generation in bilayer WSe<sub>2</sub> by layer-dependent exciton-polaron effect

**Presenter in American Physical Society March Meeting 2024** March 2024

Electric-field-switchable exciton polarons involving  $\Gamma$ -valley and K-valley Fermi sea in trilayer WSe<sub>2</sub>

## WORKING EXPERIENCE

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**Postdoctoral Researcher in University of California, Riverside** Aug 2026 to Present

Supervisors: Yong-Tao Cui and Su-Fei Shi (Co-supervised)

**Postdoctoral Research Associate in Carnegie Mellon University** Sept 2025 to Aug 2026

Supervisor: Su-Fei Shi

**Teaching Assistant in University of California, Riverside** Sept 2019 to Dec 2024

## SELECTED COURSES IN PH.D.

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- Classical Mechanics
- Electromagnetic Theory
- Quantum Mechanics
- Methods of Theoretical Physics
- Thermodynamics and Statistical Mechanics
- Condensed Matter Physics
- Nonlinear Optics

## RESEARCH SKILLS

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- **Programming:** Python
- **Sample fabrication:** Dry transfer, E-Beam Lithography, E-Beam Evaporation, Atomic Force Microscopy, Wire Bonder, Glove Box
- **Optical spectroscopy:** PL and reflectance optical setup and measurement, Montana/AttoDry2100/AttoDry1000 cryostat experience

## SELECTED PUBLICATIONS

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**arXiv:2608.01553 (2026)**

**Tianyi Ouyang**, Yuze Meng, Li Yan, Yuxuan Chen, Shuai Zhang, Xinyue Chen, Melike Erdi, Takashi Taniguchi, Kenji Watanabe, Seth Ariel Tongay, Benjamin Hunt, Ming Xie, Yong-Tao Cui, Su-Fei Shi  
**Layer-Hybridized Wigner Crystals in MoSe<sub>2</sub>/WS<sub>2</sub> moiré superlattice**

**Nature Physics accepted (2026)**

Soonyoung Cha<sup>†</sup>, Zhaoran Xu<sup>†</sup>, **Tianyi Ouyang<sup>†</sup>**, Hongyu Yao, Raj Kumar Paudel, Takashi Taniguchi, Kenji Watanabe, Andrew Y. Joe, Yia-Chung Chang, Nathaniel M. Gabor and Chun Hung Lui  
**Hybridization-revealed interlayer 2p exciton state in bilayer MoS<sub>2</sub>**

**Phys. Rev. Lett. 136 (096903) (2026) (Editor's Suggestion)**

**Tianyi Ouyang**, Erfu Liu, Soonyoung Cha, Raj Kumar Paudel, Yiyang Sun, Zhaoran Xu, Takashi Taniguchi, Kenji Watanabe, Nathaniel M. Gabor, Yia-Chung Chang and Chun Hung Lui  
**Brightening Interlayer Excitons by Electric-Field-Driven Hole Transfer in Bilayer WSe<sub>2</sub>**

**Science 390, 241-241(2025)**

**Tianyi Ouyang**, Su-Fei Shi  
**Quantum Trions in Equilibrium**

**Nano Letters 25, 4, 1459-1465 (2025)**

**Tianyi Ouyang**, Soonyoung Cha, Yiyang Sun, Takashi Taniguchi, Kenji Watanabe, Nathaniel M. Gabor and Chun Hung Lui  
**Electrically Switching Ferroelectric Order in 3R MoS<sub>2</sub> Layers**

**Nano Letters 24, 46, 14847-14853 (2024)**

Soonyoung Cha<sup>†</sup>, **Tianyi Ouyang<sup>†</sup>**, Takashi Taniguchi, Kenji Watanabe, Nathaniel M. Gabor and Chun Hung Lui  
**Enhancing Resonant Second Harmonic Generation in Bilayer WSe<sub>2</sub> by Layer-dependent Exciton-polaron Effect**

## CONTACT REFERENCE

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**Chun Hung Lui(PhD Advisor)** University of California, Riverside  
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**Sufei Shi(Postdoc Supervisor)** Carnegie Mellon University  
sufei.shi@gmail.com

**Yia-Chung Chang(Collaborator)** Research Center for Applied Sciences, Academia Sinica, Taipei  
yiachang@alum.sinica.edu.tw

**Soonyoung Cha(Collaborator)** Sookmyung Women's University  
scha@sookmyung.ac.kr